Open Project QCG, IITR

Quantum Random Walks (in Bond Pricing)

Assignment 2

Question 1

Part 1: Bond Financial Mechanics

1. Why did the corporate issuer include this 3% trigger in the bond's contract? What is their financial incentive?

The issuer is paying a huge 7% coupon every year. That coupon rate was fixed when the central bank rate was around 5%. If the central bank rate ever falls all the way to 3%, the issuer would be stuck paying 7% on the issued debt while everyone else in the market is borrowing much more cheaply. This is money the issuer is overpaying every single year. So the company builds a type of hedge against this risk. The moment rates fall to 3%, they cancel the bond, hand back the principal, and walk away. Soon after, they can go back to the market and issue fresh bonds at the new lower rate, close to 3-4%, and save a huge expense in interest over the remaining years. The trigger acts as a refinancing option for the issuer, and this is their main incentive.

2. From the investor's perspective, describe the cash flow of this bond as walking on a solid floor. What does the 3% trigger represent in this analogy, and what happens to the investor's expected cash flows the moment they step on it?

We can picture the investor walking forward in time, for example, one year at a step, collecting a 7% coupon at every step. Since we are assuming interest rates move in a straight and predictable line, the floor the investor is walking on is completely solid and flat, and there is no uncertainty about where the next step lands. The 3% trigger is a wall built on the floor at a known, fixed position. Since the path is predictable, the investor can actually calculate in advance whether that wall will ever be reached during the ten years, and if so, on exactly which day. The moment the walk reaches that wall, the floor simply ends. All future coupons are cancelled, the investor gets the principal back right there, and the walk is over.

Part 2: Stochastic Movement

1. If the current rate is 5%, and the rate takes a random walk every single day for 10 years, why is the bond more dangerous to the investor than if the rate just stayed perfectly still at 5%?

If the rate was constant at 5% forever, it would never come anywhere close to the 3% trigger, and the investor would happily collect all ten years of coupons with total certainty. However, once the rate is allowed to move randomly, it does not need to go downward on average to get close to the barrier. Even if the rate is expected to stay

around 5% on average, it can have a run of bad days and dip to 3% at some point during the ten years, and the bond will be called. Randomness creates the danger that even when the interest rate level looks safe, a random drop in the rates can erase future gains instantly.

2. How does the concept of "volatility" (the size of the daily up-and-down steps in the random walk) change the likelihood of hitting the 3% mark?

When volatility is small, the rate moves along in a narrow band near 5% and it would take an unlikely long bad streak for it to ever fall down to 3%. When volatility is large, the daily jumps are bigger, so the path spreads out much further above and below the starting point of 5% over time, and the chances of at least one of those volatility spikes dragging the rate down to the barrier rise substantially. So, higher volatility directly increases the chance of an early call of the bond, even if the expected direction of rates remains the same.

Part 3: Model Synthesis

1. How do we use the results of those 1,000 random walks to determine the fair price of this bond today? Explain conceptually how the "barrier hits" actually drag down the mathematical value of the bond.

Each of the 1,000 simulated paths represents one possible future for interest rates over the next ten years. For every path, we can work out exactly what cash the investor actually receives. In the 600 paths that never touch 3%, the investor collects all ten years of the 7% coupon plus the principal at the end. In the 400 paths that touch 3% at some point, the coupons stop the moment the barrier is hit, the investor only gets the principal back early, and misses out on every coupon that would have been received afterwards. One approach to get the fair price today is as follows: we take each path's cash flow, discount it back to present value, and then average across all 1,000 paths. We observe that 400 out of the 1,000 paths end up with a smaller total cash flow than the untouched paths, and the average value of the 1,000 paths is pulled down compared to what the bond would be worth with no barrier at all. The more paths that hit the barrier early, the more the average result gets dragged towards a smaller cash outflow, and the lower the fair value of the bond becomes today.

Question 2

Part 1: Barrier Mechanics

1. In both bonds, what is taking the random walk?

In both bonds A and B, it is the level of the stock index that is taking the random walk. The bond's coupon payments do not move on their own as they are watching the index and reacting to whichever side of the 80 barrier the index is on at any moment.

2. Assume a standard, risk-free bond with no barriers costs $100. Conceptually, why must Bond A (Down-and-Out) cost less than $100? Why does higher volatility in the stock index make Bond A even cheaper?

A standard, risk-free bond with no barriers attached is worth $100 because the buyer is guaranteed every coupon and the full principal, no matter what happens in the market. Bond A carries the same conditions, except there is now a real chance that the index dips to 80 at some time over the five years, in which case all the coupons owed in the future simply vanish. Since Bond A can only ever end up equal to or worse off than the standard bond, it must be priced lower than $100. In the higher volatility scenario, the index bounces around over a wider range, so it becomes more likely that the index will touch 80 at some point during the five years, increasing the debtor's risk much more. Hence, bond A must become cheaper.

Part 2: Defining the Walker and the Wall

1. In the Black-Cox model, what specific financial variable is taking the "random walk"? What real-world metric represents the "barrier"?

In the Black-Cox model, it is the market value of the company's assets which is taking the random walk, since the assets' valuation changes every day due to the shift in business performance, economic conditions and competition landscape. The barrier is the company's debt, equal to the liabilities it owes. The moment the stochastic value of the firm's assets falls down to or below the debt level, the company is considered to have defaulted.

2. In a simple rate-trigger bond, the barrier is a flat horizontal line (e.g., exactly 3%). But in the Black-Cox model, the default barrier often slopes upward over time. From a corporate finance perspective, why would a company's "default threshold" naturally rise as the years go by? (Hint: Think about what happens to unpaid debt or promised interest over time).

The liabilities of a company are not a fixed constant. Interest keeps compounding on any unpaid balance and it is common to take on more debt to pay the principal of a maturing bond. So, naturally the amount the company owes, counting both the original principal and increasing interest, keeps climbing the further we move forward in time. Since the barrier represents that total obligation, the barrier line slopes upward rather than staying flat. A company needs to hold onto more and more assets over time just to keep balance with what it owes, so the wall the asset walk must avoid keeps rising in sync with the growing debt burden.

Part 3: The Grand Synthesis

1. Explain how buying a standard 10-year corporate bond from Ford Motor Company is mathematically identical to buying a 10-year risk-free US Treasury bond that has a "Down-and-Out" barrier attached to it.

A risk-free US Treasury bond is treated as completely safe as it always pays every coupon and the full principal with certainty if there is no barrier attached to it. A Ford corporate bond pays the exact same coupon and principal amount, but only for as long as Ford stays solvent. If Ford's asset value ever falls down to its debt threshold, the company defaults and the future payments stop. Exactly in this manner, a down-and-out barrier US Treasury bond will get cancelled the instant the underlying assets' value touches the debt barrier. So, the Ford bond behaves like this Treasury bond in every case if the down-and-out barrier is conditional on Ford's asset value touching its debt.

2. In this analogy, what exactly is the "knock-out" event? What happens to your cash flows when the barrier is breached, and how does the Black-Cox model determine if that breach occurs?

The knock-out event here is default itself, the moment Ford's asset value first touches or falls below its debt threshold. Once that happens, the bond's future coupon payments are over, and instead of the full principal, investors typically recover only a fraction of what they are owed, depending on the assets left over. The Black-Cox model determines if this breach occurs by simulating Ford's asset value as a random walk over time and checking whether that path ever crosses the sloping debt barrier before the bond's maturity. If the path crosses at any point, that is the default date, and every future coupon payment after it is treated as lost.

3. If the market suddenly decides that Ford's business model is highly uncertain (meaning the random walk of their asset value takes wild, violent swings every day), what happens to the probability of hitting the barrier? Consequently, what must happen to the price of Ford's bond today?

Wild, violent swings in Ford's asset value imply that the random walk is exploring a much wider range of possible values than before, both far above its starting point and, more importantly, far below it. The wider spread on the downside makes it more likely that the asset value will dip down and touch the debt barrier at some point over the life of the bond, so the probability of default rises. Since default means the investor loses out on future coupons and much of the principal, volatility in the assets' value makes the bond a riskier security. Therefore, to compensate the investor for taking on that extra risk, the price of Ford's bond today must fall, which is equivalent to the yield on the bond rising.